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Bachelorthesis

Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

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Declaration of Authorship

The content of this thesis represents my own knowledge, my own understanding and my own perspective on the topic. In case artificial intelligence tools were used, their way and purpose of usage has been made transparent. Moreover, I have cited all my sources in accordance with academic standards. I am ready and able to explain and defend the positions developed in this thesis. This thesis has not been submitted, either in part or whole, at this or any other university.

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Abstract

More and more people turn to large language models for mental health support. The models respond fluently, but what drives their responses stays a black box. Ask the same therapeutic question twice and you may get two different answers. In therapy, where trust depends on predictable guidance, that variation matters. How much does it change, and at what cost?

This thesis measures response consistency in the rescripting phase of Imagery Rehearsal Therapy, where a model must choose a therapeutic strategy and carry it out. Ten models, including the EU-sovereign Mistral Large 3, two proprietary models (GPT-5.4, Claude Sonnet 4.6), and seven open-weight comparators, are evaluated across 6,000 trials at five temperature settings.

A three-level evaluation framework compares strategy selections (Jaccard similarity), response texts (BERTScore), and whether the model does what it declared (LLM judge coherence).

The main findings are: (1) At its recommended temperature, Mistral Large 3 approaches the consistency of proprietary models, with a small remaining gap. (2) Temperature affects which strategies the model picks but not how well it carries them out. The instability sits in strategy selection, not in response quality. (3) Greedy decoding ($T = 0.0$) is not optimal for all models; several achieve higher consistency at low non-zero temperatures. (4) BERTScore captures whether the therapeutic tone stays similar but does not detect changes in the underlying strategy. The three levels together reveal patterns that no single metric shows on its own.

The framework is not specific to IRT and could be applied to other structured therapy protocols.

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1. Introduction

1.1. Nightmare Disorder and Treatment

Recurring nightmares are a common clinical problem with an effective treatment: Imagery Rehearsal Therapy (IRT), a structured protocol in which the patient rescripts a nightmare under therapist guidance (Krakow & Zadra, 2006). A persistent shortage of trained therapists limits its availability (Wittchen et al., 2011), making scalable, AI-assisted delivery a practical goal.

The rescripting step, however, requires active strategy decisions from the therapist. An AI agent that selects different strategies on repeated runs of the same case is clinically unreliable, regardless of how capable any single response may be. Patients also differ in how they present: an anxious patient and a resistant patient pose different challenges, and a model that is stable for one profile may not be stable for another.

1.2. AI-Assisted Psychotherapy

Large language models are increasingly deployed in mental health, from empathy-based conversational tools to protocol-driven agents that deliver structured interventions such as cognitive behavioural therapy (Stade et al., 2024; Q. Zhang et al., 2025). reDreamAI is the deployment context for this study: a consumer-facing IRT chatbot that guides users through all stages of the protocol. It is not a research prototype but a real deployment target where reliability is non-negotiable.

This deployment context creates a specific problem. A general-purpose assistant has no therapeutic obligation. Response variability is acceptable and even desirable. A protocol-driven agent, by contrast, must make consistent strategic decisions. If the model selects different rescripting strategies for the same patient on different runs, the patient receives inconsistent care. Unlike rule-based chatbots whose responses can be reviewed once before deployment (Fitzpatrick et al., 2017), an LLM agent generates a new response on every run. Each run is a new draw from the output distribution, so stability must be evaluated repeatedly rather than verified once.

IRT targets nightmares rather than psychiatric diagnoses, which lowers the clinical risk relative to full psychotherapy. Ethical considerations are discussed in section 6.5 and section 7.2. The EU AI Act classifies autonomous therapeutic AI as high-risk and requires auditability (Aboy et al., 2024; van Kolfschooten & van Oirschot, 2024). Data sovereignty requirements push toward EU-resident models (Dale, 2025), introducing a constraint on model selection discussed in section 3.8.

1.3. The Evaluation Gap

Standard Natural Language Processing (NLP) benchmarks measure what a model *can* do, not whether it does the same thing reliably. A model can score well on a single evaluation while producing different therapeutic strategies on repeated runs of the same input. Capability and stability are distinct properties, but current benchmarks conflate them.

Prior therapeutic AI evaluations, including the closest existing work (BOLT, Chiu et al., 2024), measure whether a model applies appropriate techniques. They evaluate single responses rather than the distribution across repeated runs (Talat et al., 2024), so they cannot detect cross-run inconsistency (section 2.2). Systematic, reproducible in silico trials offer a way to measure stability at scale without requiring human participants.

1.4. Research Objectives

This thesis addresses the following research question:

> How consistent are LLM therapeutic responses across independent runs when given the same patient context within a structured IRT protocol?

Two contributions follow from this question.

Contribution 1. A three-level evaluation framework that measures stability at each layer: *plan consistency* (do strategy selections repeat?), *response similarity* (do the generated texts convey the same meaning?), and *plan-response coherence* (does the model do what it declared?). Together, these layers reveal divergence patterns that no single metric can detect.

Contribution 2. A cross-model comparison that applies this framework to sovereign, proprietary, and open-weight models under controlled temperature conditions.

The study isolates the IRT rescripting phase because it is the stage that requires active strategy decisions. Isolating a single stage eliminates stage-routing confounds and allows any measured instability to be attributed to the therapist model rather than to upstream variation in the conversation. chapter 3 describes the experimental design in detail.

2. Background

2.1. Imagery Rehearsal Therapy

Nightmare disorder affects an estimated 2–5% of the adult population (Li et al., 2010; Sandman et al., 2013; Schredl, 2010) and shows strong comorbidity with Post-Traumatic Stress Disorder (PTSD) and depression (Morgenthaler et al., 2018). For patients with recurrent nightmares, the distress extends beyond the dream itself: sleep avoidance, daytime fatigue, and heightened anxiety create a cycle that standard sleep hygiene cannot break. IRT is the recommended treatment for this condition (Krakow & Zadra, 2006; Morgenthaler et al., 2018).

IRT follows a structured three-stage protocol. In the *recording* stage, the patient describes the nightmare in detail. In the *rescripting* stage, the patient and therapist collaboratively transform the nightmare narrative to reduce distress and increase mastery. In the *rehearsal* stage, the new narrative is consolidated and the patient prepares to rehearse it before sleep. Krakow and Zadra (2006) describe the rescripting instruction as deliberately open-ended: patients are told to "change the nightmare any way you wish."

The rescripting stage is the cognitive core of the protocol. Unlike recording (which requires listening) or rehearsal (which requires repetition), rescripting demands that the therapist choose a specific rescripting strategy and guide the patient through it. These choices determine the character of the intervention. A therapist who helps the patient confront a nightmare threat delivers a different treatment from one who introduces a protective figure or reframes the threat as harmless.

Clinical research has identified distinct categories of rescripting strategies. Harb et al. (2012) coded five types from combat veteran rescripts: alternative endings (58%), positive image insertion (23%), threat transformation (13%), reality markers (10%), and distancing (8%). Germain et al. (2004) proposed a complementary classification through the Multidimensional Mastery Scale, which groups rescripting outcomes along physical, social, environmental, emotional, mystical, and avoidance dimensions. Both frameworks show that rescripting is not a single technique but a family of distinct interventions. Which strategy the therapist selects is therefore a meaningful clinical decision, and the consistency of that decision across sessions is what this thesis measures.

Treatment fidelity instruments provide the clinical standard for measuring whether a therapist delivers an intervention correctly. The IRT literature draws on two established frameworks: the ENACT rating scale (Kohrt et al., 2015) and the NIH BCC recommendations (Bellg et al., 2004). Both grade therapist adherence on a ternary scale: *absent*, *partial*, or *implemented*. This gradation captures clinically meaningful differences. A therapist who briefly mentions a rescripting strategy without developing it (partial) delivers a different intervention from one who fully guides the patient through it (implemented).

This ternary framework provides a scoring foundation that transfers directly to computational evaluation (see subsection 3.7.3). However, existing fidelity instruments assume a human clinician. They have not been adapted for Large Language Model (LLM)-based agents, where the therapist is a probabilistic text generator rather than a trained professional.

2.2. Treatment Fidelity in LLM-Based Interventions

The deployment of LLMs in mental health has expanded from empathy tools and symptom screeners to protocol-driven therapeutic agents. Stade et al. (2024) survey this trajectory and argue that LLMs could fundamentally reshape behavioural healthcare delivery, from diagnostic support to full psychotherapeutic interventions. Early systems such as WOEBOT (Fitzpatrick et al., 2017) delivered Cognitive Behavioral Therapy (CBT) through rule-based dialogue trees where every response was pre-authored and reviewed before deployment. For these systems, treatment fidelity is a design property: the chatbot cannot deviate from its script.

LLM-based agents operate differently. Each response is a fresh sample from the model's output distribution. A single model can produce high-quality, protocol-adherent responses on one run and strategically inconsistent ones on the next. This distinction is central to the treatment fidelity literature. Mowbray et al. (2003) define fidelity as the extent to which an intervention follows its intended protocol, covering dimensions such as adherence (doing the prescribed things) and quality of delivery (doing them consistently). For rule-based chatbots, adherence is built in by construction. For LLM-based protocol agents, adherence becomes probabilistic and must be measured across runs rather than assumed from a single review.

LLM-based agents also face general failure modes such as hallucination, stage confusion, and persona inconsistency. These are well-documented concerns for clinical AI but fall outside the scope of this study. The present work focuses on a narrower problem: the probabilistic nature of LLM outputs means that identical therapeutic contexts can produce different outputs on each run. Three types of inconsistency can occur, each with different clinical implications.

First, the model may select different rescripting strategies across runs. As established in section 2.1, strategy selection is a meaningful clinical decision. A patient who receives confrontation-based rescripting in one session and cognitive reframing in the next encounters two different interventions. In a protocol where the therapist is expected to follow consistent principles, this is a fidelity problem in the sense of Mowbray et al. (2003): the model fails to adhere to a stable treatment approach.

Second, even when the model selects the same strategy, the therapeutic language may vary between runs. Some variation is expected and acceptable: a competent human therapist naturally paraphrases while maintaining the same strategic intent. But large variation in wording, even under the same strategy, could indicate unstable delivery.

Mowbray's fidelity framework motivates this concern: if consistent delivery matters for human therapists, it matters at least as much for LLM agents whose outputs vary with every run. This study does not measure fidelity to a clinical standard directly. Instead, it measures *consistency across runs*: does the model plan the same techniques each time (Method 1), does it word the response similarly (Method 2), and does it carry out what it declared (Method 3).

Third, the model may declare one strategy but carry out another. If a model states that it will use confrontation but actually delivers a calming sensory exercise, there is a mismatch between intent and execution. This internal incoherence is distinct from the first two types: the model is neither selecting a wrong strategy nor varying its language, but failing to do what it said it would do.

Each type of inconsistency requires a different measurement approach. chapter 3 develops one method for each.

The European Union (EU) AI Act and Medical Device Regulation both require auditability for software used in clinical contexts (section 1.2), yet the type of stability described above remains largely unmeasured. Prior therapeutic AI evaluation is qualitative, post-hoc, or single-session. Chiu et al. (2024) represent the closest prior work with their BOLT framework, which assesses LLM therapist behaviour across 13 behavioural codes drawn from motivational interviewing. BOLT asks whether the model does the right thing. The present study asks a different question: whether the model does the *same* thing across repeated runs. Existing frameworks, including BOLT, evaluate single responses rather than the distribution of responses across repeated trials.

2.3. Stochastic Behaviour in Language Models

At temperatures above zero, LLM token selection is probabilistic. The softmax distribution over the vocabulary is divided by the temperature parameter before sampling, so higher temperatures flatten the distribution and lower temperatures sharpen it toward the most likely token. At $T = 0$, the model selects the highest-probability token deterministically (greedy decoding). At $T > 0$, tokens are sampled from the resulting distribution, introducing variance into the output.

This per-token variance compounds through autoregressive generation. Each token is conditioned on all preceding tokens, so a single divergent token early in a response can cascade into a different therapeutic strategy. A model that selects "confront" rather than "imagine" at one token position may generate an entirely different rescripting approach in the subsequent response. The clinical impact of stochastic sampling is therefore not proportional to per-token entropy but can amplify through the response.

The clinical consequence follows directly: the same patient presenting the same nightmare on two different days may receive strategically inconsistent care. In a general-purpose assistant, this variability is benign or even desirable. In a structured therapy protocol like IRT, where the therapist is expected to apply consistent techniques across sessions, it maps directly to the three types of inconsistency described in section 2.2: variable strategy selection, variable response wording, and plan-response mismatch.

Prior work on LLM stability addresses different questions than therapeutic consistency. Novikova et al. (2025) survey the consistency landscape, covering whether models give the same answer to the same question across different phrasings and contexts. Xu et al. (2025) measure the opposite problem: LLMs generating repetitive plot structures in creative writing, producing too little variation rather than too much. Blackwell et al. (2024) focus on benchmark reproducibility, showing that evaluation scores fluctuate across repeated runs even with fixed seeds and $T = 0.0$.

These lines of research share the concern for cross-run variation but none of them ask whether a model makes the same *clinical decisions* when presented with identical patient contexts.

The mechanical source of instability is well understood. What remains unquantified is its impact on structured therapeutic decision-making. How much does temperature-driven variance cost in clinical consistency? This question motivates the temperature sweep in the experimental design (section 3.9), which tests five points from $T = 0.0$ (deterministic) to $T = 0.6$ (typical vendor default for open-weight models).

2.4. Evaluation Metrics for Text Generation

Measuring consistency between text outputs requires a similarity metric. The choice of metric determines what kind of consistency is captured, and standard options fail in different ways for therapeutic language.

Surface-level metrics such as BLEU and ROUGE work by counting how many short word sequences (n-grams) appear in both texts, producing a score between 0 and 1.

In therapeutic contexts, token overlap is misleading. A therapist naturally varies their wording while keeping the same clinical intent, so two responses that implement the same strategy through different phrasing score low on BLEU/ROUGE despite being therapeutically equivalent.

BERTScore (T. Zhang et al., 2020) avoids this problem by comparing texts in contextual embedding space rather than at the token level. It measures whether two texts convey the same meaning regardless of surface wording. The embedding model used in this study, DeBERTa-XLarge-MNLI (He et al., 2021), achieves the highest correlation with human similarity judgements ($r = 0.778$) among available models. No validation specific to therapeutic language exists, so this study uses its general-domain performance as the best available approximation (subsection 3.7.2).

BERTScore captures semantic similarity but cannot assess strategic intent. A metric that reports high similarity between two responses does not reveal whether both responses implement the same therapeutic strategy. The LLM-as-judge paradigm addresses this gap. By instructing a separate LLM to evaluate whether a response implements a declared strategy, it becomes possible to assess criteria that no scalar metric can capture: “does this response implement a confrontation strategy?" requires understanding both the strategy definition and the response content.

LLM-as-judge approaches carry known limitations. The judge's scores are not ground truth but a computational estimate. The judge model brings its own biases: self-preference (rating outputs from its own family more favourably), position bias (higher scores for responses presented first), and verbosity bias (rewarding longer responses regardless of quality) (Zheng et al., 2023). These can be mitigated through cross-model judging, structured rubrics, and explicit justification requirements, but they cannot be eliminated entirely. Any study using an LLM judge must treat its scores as approximate and, where possible, validate them against human ratings. The mitigation and validation strategy adopted in this study is detailed in subsection 3.7.3.

No single metric covers what therapeutic stability requires. Embedding similarity captures semantic similarity but misses strategic intent. LLM judges can assess intent but introduce their own reliability concerns. Surface metrics capture neither. Existing therapeutic AI evaluations rely on a single metric type, leaving the gap between semantic equivalence and strategic intent unaddressed. This motivates the three-method design presented in chapter 3.

3. Methods

3.1. System Overview and Design Rationale

The evaluation system separates dialogue creation from stability measurement through two decoupled stacks. The *generation stack* produces synthetic therapy sessions by running a multi-agent dialogue loop. The *evaluation stack* measures the stability of therapeutic responses by presenting frozen conversation histories to the model under test and collecting multiple independent outputs.

This separation is a deliberate design choice. If the evaluation stack generated its own dialogue context, any measured instability could reflect upstream randomness in the patient or routing agents rather than genuine variation in the therapist model. By freezing the conversation history before evaluation begins, the system ensures that every trial receives identical input. Any observed inconsistency is attributable solely to the model under test.

The end-to-end data flow proceeds in three stages: patient vignettes feed the generation stack, which produces complete therapy sessions. These sessions are sliced at rescripting-turn boundaries to create frozen histories. The evaluation stack presents each frozen history to the therapist model across multiple independent trials, collecting both a strategy declaration and a therapeutic response per trial. Finally, an aggregation pipeline aligns all metrics across conditions for comparative analysis.

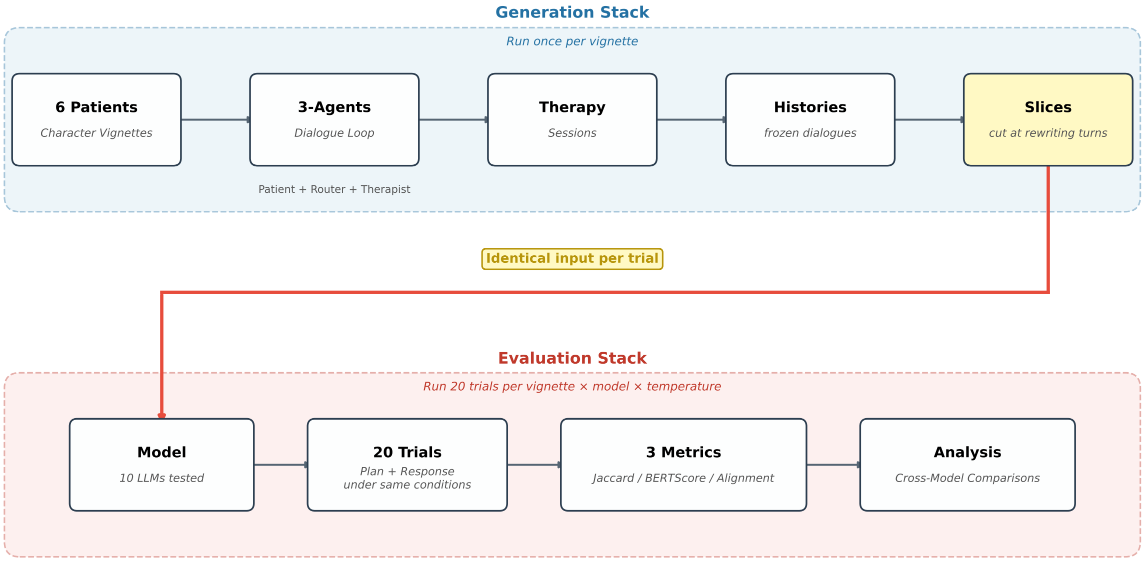

Figure 3.1.: Two-stack pipeline. The generation stack produces frozen conversation histories (run once). The evaluation stack presents each history to the model under test across twenty independent trials per condition. The stacks are decoupled so that evaluation-side variance cannot be contaminated by generation-side randomness.

3.2. Dialogue Generation

The generation stack orchestrates a three-agent loop that produces naturalistic IRT sessions. Each turn passes through three agents:

- **Patient:** responds based on a behavioural profile (nightmare content, emotional style, engagement pattern).
- **Router:** classifies the current protocol stage (recording, rescripting, or rehearsal).
- **Therapist:** generates a stage-appropriate response, guided by the router's classification.

The vignette design is informed by anonymised session data from a prior reDreamAI efficacy study. Because data protection requirements prevent using real patient transcripts directly, synthetic patient profiles were created and validated against published clinical research on simulated patients.

Six vignettes cover the range of clinical presentation difficulty expected in IRT deployment: *anxious*, *avoidant*, *cooperative*, *resistant*, *skeptic*, and *trauma*. These profiles range from a cooperative patient who engages readily with rescripting instructions to a trauma patient whose nightmare content triggers intense emotional responses during the session. The design draws on prior synthetic patient work. Z. Wang et al. (2024) demonstrate that expert-defined behavioural principles produce clinically plausible simulated patients. R. Wang et al. (2024) provide cognitive conceptualisation diagrams and conversational styles (direct, resistant, expressive, minimal, deviating, agreeable) that parallel the six profiles used here. Six profiles sample the difficulty range but do not claim exhaustive coverage of patient variability (see section 7.2).

The generation stack traverses the full three-stage IRT protocol, producing complete therapy sessions that are then sliced to create evaluation entry points. Figure 3.2 illustrates this process: the dialogue loops through recording and rescripting until each stage is complete, then proceeds to rehearsal. Frozen history slices are cut from the rescripting stage.

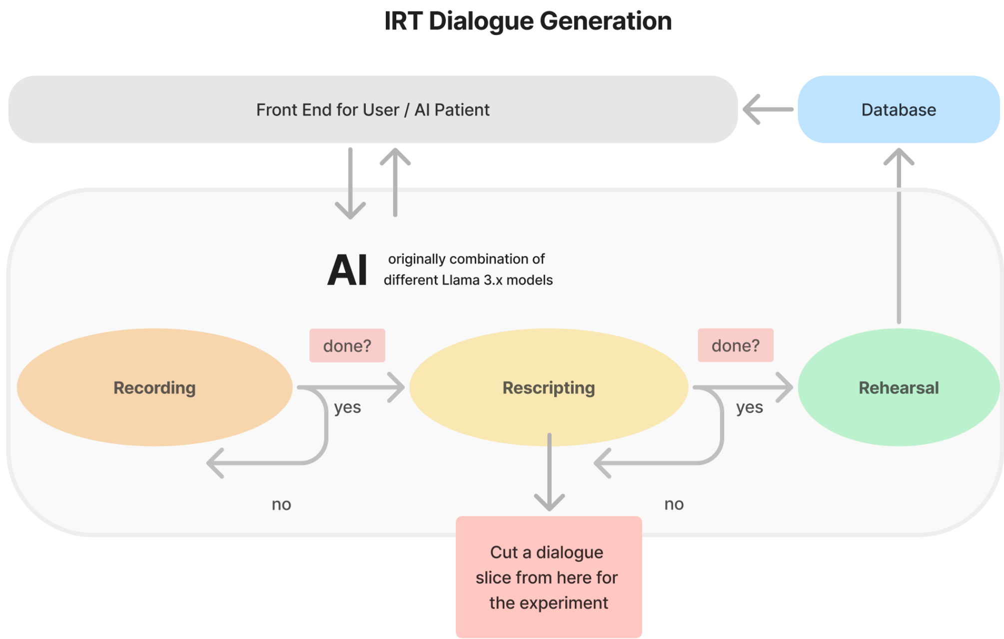

Figure 3.2.: Dialogue generation loop. The three-agent system traverses recording, rescripting, and rehearsal. Frozen history slices for the experiment are cut at rescripting-turn boundaries.

3.3. Frozen History Design

Frozen conversation histories serve as deterministic evaluation entry points. Each vignette's complete therapy session is sliced at rescripting-turn boundaries, where each slice ends after a complete therapist-patient exchange. Every trial, across all models and temperatures, receives the identical frozen history as input. This design eliminates context variation as a confound: if two trials produce different therapeutic responses, the difference is attributable to the model's stochastic sampling, not to differences in the input conversation.

A preliminary analysis tested Mistral Large 3 at five slice depths across all six vignettes at $T = 0.075$ and $T = 0.15$. Jaccard consistency was lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rose to near-ceiling values from the third slice onward ($J > 0.93$). The main experiment evaluates at the second rescripting-turn boundary (`slice_2`), making the measured stability a conservative estimate. Slice depth effects are reported in section 5.6.

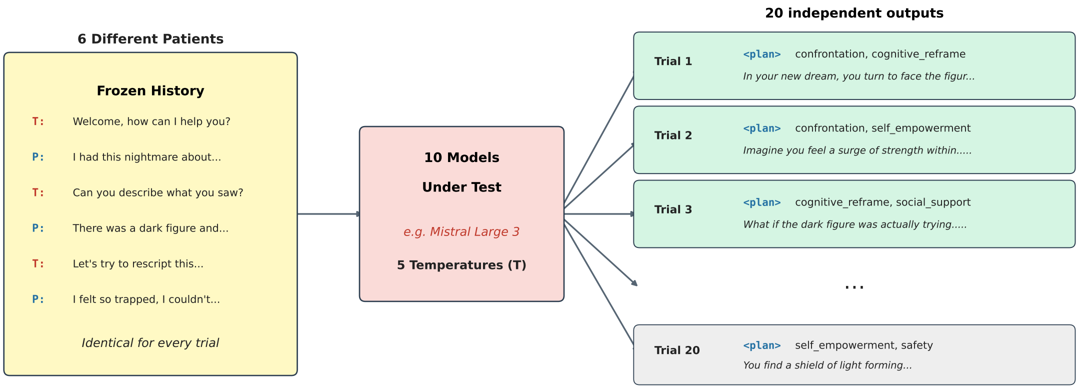

Figure 3.3.: Measurement design. The same frozen history is presented to the model under test twenty times. Each trial independently produces a plan and a therapeutic response. Variation across trials is the instability this study measures.

3.4. Evaluation Stack

The evaluation stack bypasses the router and injects the rescripting prompt directly into the therapist model. Because the IRT stage is known a priori (all frozen histories are sliced within the rescripting phase), routing is unnecessary and would introduce an additional source of variation. Bypassing it isolates the therapist model from stage-classification error.

Each evaluation trial uses *fused generation*: the model produces a `<plan>` block declaring one to two therapeutic strategies, followed by the therapeutic response, in a single call. The plan tokens condition the response tokens in a Chain-of-Thought (CoT)-style generation, so the response is directly conditioned on the declared strategy. This fused approach enables the coherence analysis in Method 3 (subsection 3.7.3): because plan and response are generated in a single pass, any mismatch between declared intent and executed strategy reflects the model's own inconsistency rather than a decoupling artefact. Figure 3.3 illustrates this measurement design.

Twenty independent trials are collected per condition, yielding $\binom{20}{2} = 190$ pairwise comparisons. This trial count balances statistical power with compute cost: 190 pairs provide robust coverage of the pairwise similarity distribution while keeping the total experiment within budget.

3.5. The Plan Mechanism

The `<plan>` block raises a methodological question. The faithfulness literature shows that CoT explanations do not always reflect a model's actual computation (Lanham et al., 2023; Turpin et al., 2023). A declared therapeutic strategy may therefore be a post-hoc rationalisation rather than a window into the model's decision-making. This study sidesteps the problem by framing the plan as *declared intent*: the plan captures what the model commits to, not what it internally computes. Whether or not the declaration reflects genuine reasoning, a practical consequence holds: if strategy declarations vary across runs, clinical responses will also vary.

This framing sidesteps the faithfulness debate entirely while preserving three uses. First, the declared strategies enable consistency scoring on strategy level (Method 1, subsection 3.7.1). Second, comparison between declared strategies and the executed response enables coherence verification (Method 3, subsection 3.7.3). Third, the plan provides a human-readable summary of the model's intended approach, supporting clinical oversight in deployment.

3.6. Strategy Taxonomy Development

A fixed taxonomy constrains the `<plan>` block to a predefined set of therapeutic strategies. This constraint is necessary for quantitative evaluation: free-text plans cannot be aggregated into set-similarity scores. The taxonomy evolved through four iterations to resolve distribution skew in early versions (full revision history in Appendix A). The core design principle is that each category must describe a rescripting *mechanism* rather than a therapeutic *goal*. Early versions that used goal-level labels (e.g. *empowerment*, *mastery*) produced extreme skew because every rescripting intervention serves the goal of increased mastery. Splitting categories along the mechanism dimension resolved this.

The final six-category taxonomy (Table 3.1) captures concrete therapeutic mechanisms:

The six categories operationalise the clinical rescripting strategies introduced in section 2.1. Avoidance and distancing are excluded because they do not resolve the nightmare threat. Albanese et al. (2022) review that IRT works by increasing mastery over the nightmare, and that avoidance reinforces the fear memory rather than overwriting it. Harb et al. (2012) found that distancing was the least-used

Table 3.1.: Strategy taxonomy (v4). Each category describes a rescripting mechanism rather than a therapeutic goal.

Category	Mechanism
confrontation	The dreamer directly faces, fights, or stands up to the threat
self-empowerment	The dreamer gains a new ability or inner strength
safety	A physical barrier, shelter, or protective figure shields the dreamer
cognitive reframe	Existing dream elements change meaning
social support	A new person, companion, or ally is introduced
sensory modulation	Sensory qualities shift, including internal somatic calming

strategy (8%) and that violence in rescripted dreams predicted worse outcomes. Reality markers fall outside single-turn rescripting.

Taxonomy design directly shapes measured consistency. A coarse taxonomy inflates agreement (every trial lands in the same category), a fine one deflates it. The sensitivity of scores to category boundaries is revisited in the discussion (section 6.5).

3.7. Evaluation Metrics

Three evaluation methods target distinct layers of the generation process. No single metric captures the full picture of therapeutic stability (section 2.4). Method 1 asks whether the model makes consistent therapeutic decisions. Method 2 asks whether it produces consistent therapeutic responses. Method 3 asks whether it does what it says it will do.

3.7.1. Method 1: Plan Consistency

Method 1 measures whether the model selects the same therapeutic strategies across independent trials by extracting strategy sets from the `<plan>` block of each trial and comparing them with pairwise Jaccard similarity:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{3.1}$$

The mean Jaccard score is computed over all $\binom{20}{2} = 190$ trial pairs per condition. Jaccard is appropriate because strategy combinations are unordered sets: a model that declares *confrontation / cognitive reframe* has made the same clinical decision as one that declares *cognitive reframe / confrontation*.

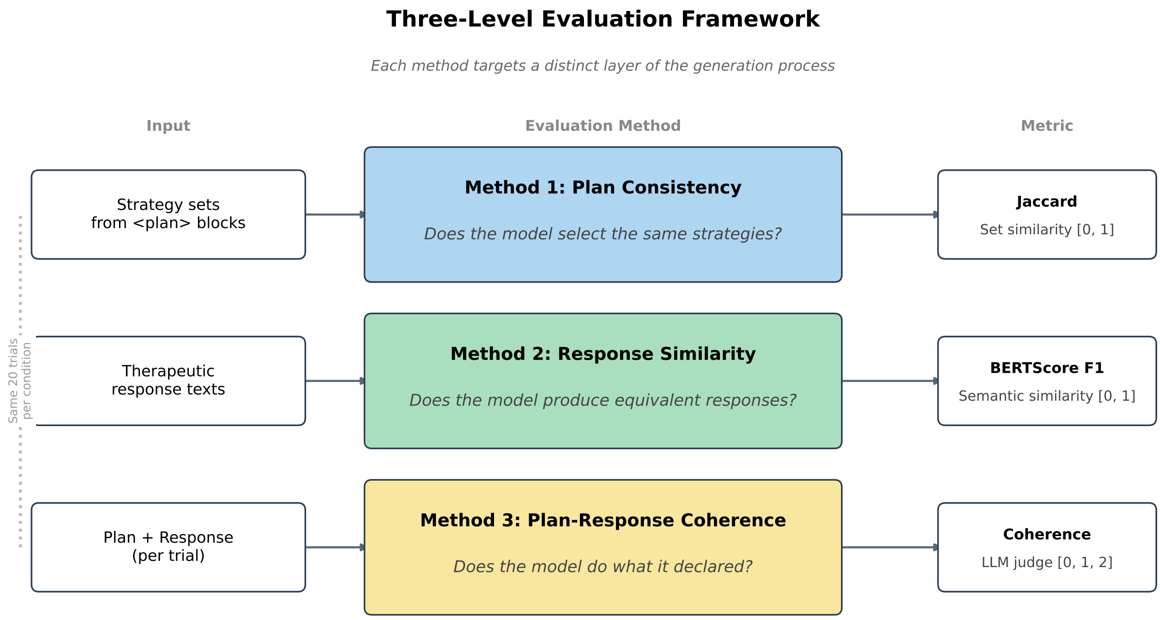

Figure 3.4.: Three-level evaluation framework. Each method targets a distinct layer of the generation process: strategy selection (Jaccard), response wording (BERTScore), and plan-response coherence (LLM judge).

Because Jaccard operates on discrete sets of one to two items drawn from six categories, it takes only a small number of distinct values: $\{0, 0.33, 0.5, 0.67, 1.0\}$. All three metrics are reported with means and standard deviations, aggregated across vignettes.

Plan validity rate serves as an upstream quality gate. A malformed `<plan>` block that cannot be parsed into valid strategy identifiers is excluded from the Jaccard computation. The validity rate reports the proportion of trials with parseable plans, indicating whether the model can follow the structured output format at all.

3.7.2. Method 2: Response Similarity

Method 2 measures whether the model produces semantically similar responses across trials, independent of strategy declarations. Pairwise BERTScore F1 is computed over response texts (with plan blocks excluded) across the same 190 trial pairs. The embedding model is DeBERTa-XLarge-MNLI (He et al., 2021), selected for its high correlation with human similarity judgements as discussed in section 2.4.

3.7.3. Method 3: Plan-Response Coherence

Method 3 measures whether the model's therapeutic response implements its declared strategy. A separate LLM judge scores each declared strategy on a ternary scale:

- **0 (absent):** The strategy is not reflected in the response.
- **1 (partial):** The strategy is mentioned but not fully developed.
- **2 (implemented):** The strategy is clearly and substantively present.

This ternary scale follows the ENACT and NIH BCC fidelity frameworks introduced in section 2.1.

The judge model is GEMINI 3.1 PRO, accessed through Google AI Studio at $T = 1.0$, the Gemini 3 default. Lowering the temperature below 1.0 can cause looping or degraded output for Gemini 3 models (Google, 2026). GEMINI is from a different model family than all evaluation targets, preventing self-preference bias. Its thinking mode supports structured reasoning about whether a response implements a given strategy. The judge receives the strategy definitions, the declared strategies, and the response text, and produces a score with justification for each strategy. GEMINI 3 PRO has been validated as a reliable judge for content validity assessment in psychology (Poźniak et al., 2026).

Bias mitigations follow established recommendations (Zheng et al., 2023): cross-model judging, CoT justification, and a transparent rubric anchoring scores to specific behavioural criteria. Natural Language Inference (NLI) cross-encoders were considered but rejected after preliminary testing showed an F1 ceiling of approximately 0.55 on this task.

Judge reliability is validated through human annotation of a stratified random subset of 60 trials (covering all ten models), scoring the same ternary rubric. Cohen's κ between human and judge is 0.777 (linear-weighted), indicating substantial agreement. Exact agreement is 89.2%. All 13 disagreements go in one direction: the human annotator scored higher than the judge, indicating a conservative rather than lenient bias.

3.8. Model Selection

EU data sovereignty serves as the primary selection criterion for the study's main subject. The deployment context (reDreamAI) requires a model that can be selfhosted within EU data residency boundaries (Dale, 2025). All evaluation targets run in non-thinking mode for fair comparison.

The primary subject is Mistral Large 3 (675B Mixture of Experts (MoE), 41B active parameters, Apache 2.0), the EU-sovereign model that reDreamAI would deploy in production. Two additional Mistral models complete the family comparison: Mistral Small 4 (119B MoE, 6.5B active) is the newest and most parameter-efficient MoE in the set, and Mistral Small 3.2 (24B dense) serves as a dense predecessor baseline.

Two proprietary models provide a ceiling reference. GPT-5.4 (non-thinking variant) is the strongest proprietary model without thinking overhead. Claude Sonnet 4.6 adds a second proprietary reference point from a different vendor.

Five open-weight comparators span three size classes and two architectures. Table 3.2 lists all ten models with their selection rationale.

The selection provides size-class diversity (24B–675B), architectural diversity (dense vs. MoE), and provider diversity across five vendors. Qwen 3.5 and Mistral Small 4 are hybrid-thinking models that support optional reasoning. To maintain fair comparison, reasoning is disabled for all evaluation targets. All OpenRouter calls specify the model vendor as the inference provider, ensuring that each model runs on the same backend across all trials.

Including three Qwen scales (27B, 122B, 397B) enables within-family scaling analysis.[1]

[1] DeepSeek V3.2 was originally considered for this slot but dropped because its internal temperature scaling would have complicated cross-model comparison.

Table 3.2.: Evaluation targets. All models run in non-thinking mode via OpenRouter.

Group	Model	Size	Selection rationale
EU-sovereign (Mistral, Apache 2.0)			
Primary	Mistral Large 3	675B MoE (41B active)	EU-sovereign flagship. Production deployment candidate
EU	Mistral Small 4	119B MoE (6.5B active)	Newest Mistral. Fewest active parameters of any MoE in set
EU	Mistral Small 3.2	24B dense	Dense predecessor to Small 4
Open-weight comparators			
MoE	Qwen 3.5 397B	397B MoE (17B active)	Large MoE comparator to Mistral Large 3
MoE	Qwen 3.5 122B	122B MoE (10B active)	Peer MoE comparator to Mistral Small 4
Dense	Qwen 3.5 27B	27B dense	Same size class as Mistral Small 3.2
Dense	OLMo 3.1 32B	32B dense	Fully open (weights, data, code). Best provenance
Dense	Llama 3.3 70B	70B dense	Continuity with prior efficacy study
Proprietary ceiling			
Ceiling	Claude Sonnet 4.6	Proprietary	Latest Claude model, known for natural conversational style
Ceiling	GPT-5.4	Proprietary	Strongest non-thinking proprietary model

3.9. Experimental Conditions

The experiment follows a full factorial design:

Rather than testing only two temperature conditions, the experiment uses a five-point temperature sweep that spans from deterministic decoding to the vendor-recommended defaults. The scale is anchored at three points motivated by vendor documentation:

$T = 0.0$ (greedy decoding) serves as a deterministic upper bound on stability. Even at $T = 0.0$, GPU parallelisation introduces floating-point non-determinism (Yuan et al., 2025), so the twenty-trial design captures residual variance directly (Blair-Stanek & Van Durme, 2025).

$T = 0.15$ is the recommended temperature for Mistral models. Mistral Small 3.2 sets this value explicitly in its HuggingFace generation configuration. Mistral Large 3 and Mistral Small 4 lack a generation configuration default. Their model cards recommend temperatures below 0.1 for production use, while code examples on the same cards use $T = 0.15$. Because no authoritative single default exists for these two models, 0.15 was chosen to match the verified Small 3.2 value and the vendor code examples.

$T = 0.6$ is the default for Qwen 3.5 and OLMo 3.1, also close to the standard Llama chat range (0.6–0.7) and the proprietary defaults (GPT-5.4 and Sonnet 4.6 default to $T = 0.7$).

The intermediate points $T = 0.075$ and $T = 0.3$ fill the gap between Mistral's low default and the Qwen/Llama range. $T = 0.075$ was added after an initial four-point run revealed that several models peak in consistency between $T = 0.0$ and $T = 0.15$, suggesting that a small amount of sampling variance can be beneficial (discussed in section 5.2).

Table 3.3.: Experimental conditions. Total: $6 \times 5 \times 20 = 600$ trials per model, 6,000 across all ten models.

Dimension	Values
Vignettes	anxious, avoidant, cooperative, resistant, skeptic, trauma
Trials	20 per condition
Temperatures	$T \in \{0.0,\ 0.075,\ 0.15,\ 0.3,\ 0.6\}$
Models	10 evaluation targets (Table 3.2)

3.10. Statistical Analysis

The analysis is primarily descriptive. For each model and temperature condition, central tendency and dispersion are reported for all three evaluation metrics, aggregated across vignettes. All three metrics are reported with means and standard deviations.

Spearman rank correlations serve two purposes. First, per-model correlations between Jaccard and temperature quantify the temperature sensitivity of each model across the five-point sweep. Second, cross-method correlations (Method 1 vs. Method 2, Method 1 vs. Method 3, Method 2 vs. Method 3) assess whether the three evaluation layers capture redundant or independent information. If they diverge, the clinical implications differ: high plan consistency with low response similarity indicates stable strategy selections but variable execution, while the reverse indicates different strategies producing similar surface outputs. Spearman rather than Pearson is used because Jaccard takes only a small set of discrete values and BERTScore distributions may be non-normal.

Kruskal-Wallis tests assess whether models and vignettes differ significantly on each metric. These are exploratory rather than confirmatory: the study is designed to describe stability patterns, not to test a preregistered hypothesis.

No fixed threshold defines “stable enough.” Clinical significance thresholds for LLM stability do not yet exist in the literature. Results are interpreted comparatively: model against model, across the temperature sweep. The random Jaccard baseline (expected value for random 2-of-6 selections, approximately 0.2) provides a lower reference point.

4. Implementation

This chapter describes how the methods from chapter 3 are realised in software. The pipeline is implemented in Python and structured around three concerns: a provider-agnostic LLM interface, a config-driven experiment runner, and a post-hoc aggregation pipeline. All source code is available in the project repository[1].

4.1. Software Architecture

The pipeline follows an async-first design. Every LLM call is non-blocking, and trials within a condition run concurrently via `asyncio.gather()`. This is a direct consequence of the factorial design: 600 trials per model would be prohibitive if executed sequentially. Parallel execution reduces wall-clock time from hours to minutes per model.

Configuration: Two YAML files define the full experiment declaratively. `models.yaml` specifies provider endpoints, model identifiers, role assignments, and evaluation targets. `experiment.yaml` controls sampling parameters (trial count, temperatures, maximum token length), metric selection, and output paths. Changing the evaluation target or temperature requires editing a single YAML field rather than modifying code. This separation ensures that experiment conditions are reproducible: the configuration snapshot saved with each run fully specifies how to recreate it.

Data validation: Pydantic models enforce type safety at every data boundary. The `Message` and `Conversation` classes validate all fields on construction and serialisation. When a frozen history is loaded from disk, `Conversation.from_dict()` uses `model_validate()` with `extra="ignore"`, silently discarding metadata fields that do not belong to the conversation schema. This is intentional: frozen histories may

[1] https://github.com/me-daniel/measure-ai-drift

Table 4.1.: Package structure. Each package maps to a distinct concern in the pipeline.

Package	Responsibility
`src/agents/`	Agent implementations (patient, therapist, router). Each agent extends a `BaseAgent` class that handles message formatting, system prompt injection, and provider delegation
`src/core/`	Data models (`Message`, `Conversation`), stage and language enums, YAML and prompt loaders
`src/llm/`	Provider abstraction. Single-class interface to all LLM backends
`src/stacks/`	Orchestration. `GenerationStack` runs the three-agent dialogue loop. `EvaluationStack` runs fused plan-and-response generation from frozen histories
`src/evaluation/`	Experiment runner, multi-trial sampler, and all three evaluation metrics (Jaccard, BERTScore, LLM judge coherence)

carry generation-time metadata (timestamps, model versions) that is irrelevant to evaluation.

The same validation applies to plan extraction. The `extract_plan_strategies()` function parses the `<plan>` block from each trial output and validates each strategy identifier against the taxonomy. A malformed plan that cannot be parsed into valid strategy identifiers fails at extraction time rather than propagating downstream where it would silently corrupt Jaccard scores.

Provider abstraction: A single `LLMProvider` class wraps an `AsyncOpenAI` client and exposes one async method: `generate(messages, **kwargs)`, returning a tuple of response content and token usage. All ten evaluation targets, the patient agent, and the router use OpenRouter through this interface. The Gemini judge uses Google AI Studio through the same interface, since Google AI Studio exposes an OpenAI-compatible endpoint. Adding a new model or provider requires a configuration entry, not a code change.

A factory function (`create_provider()`) resolves role names (e. g. "therapist", "judge") to concrete provider configurations by looking up the active model assignment in `models.yaml`. For experiment-tier runs, the judge automatically upgrades from Gemini Flash Lite (used during development) to Gemini 3.1 Pro.

The codebase is organised into five packages:

4.2. Experiment Execution

An `ExperimentRun` object manages a single evaluation run: one model, one vignette, one temperature, twenty trials. On initialisation it creates a timestamped output directory and saves the configuration snapshot and frozen history.

Execution proceeds in three phases:

1. **Sampling.** A `Sampler` wraps the `EvaluationStack` and runs twenty independent trials. Each trial calls the therapist model once (fused mode), producing a `<plan>` block and a therapeutic response. Trials run in parallel via `asyncio.gather()` with built-in retry logic for rate-limit errors (HTTP 429) using exponential backoff.

2. **Metric computation.** After all trials complete, the run computes all three evaluation metrics. Method 1 (Jaccard) and Method 2 (BERTScore) are computed locally. Method 3 (coherence) issues parallel async calls to the Gemini judge, with a 120-second timeout per trial to handle unresponsive judge calls gracefully.

3. **Persistence.** Results are saved to the run directory.

Each run produces a self-contained output bundle:

```
experiments/runs/{timestamp}_{model}_{vignette}/
    config.yaml            # run parameters (model, temp, slice)
    frozen_history.json    # input conversation (irreplaceable)
    trials/
        trial_01.json      # plan + response + token usage
        ...
        trial_20.json
    metrics.json           # Jaccard, BERTScore, validity, coherence
    judgments.json          # raw judge outputs with reasoning
```

The frozen history is the only irreplaceable artefact in this structure. If the evaluation logic changes (a bug fix in Jaccard computation, a new judge prompt), metrics and judgements can be recomputed from the saved trials. Trials themselves can be recomputed from the frozen history plus the configuration, though at the cost of additional Application Programming Interface (API) calls.

The plan extraction logic handles several output formats that different models produce. The primary pattern matches a closed `<plan>` tag. Fallback patterns handle unclosed tags, bare tag-style strategy names, and plain-text first-line declarations. This robustness is necessary because the ten evaluation targets do not all follow the structured output format identically, despite receiving the same system prompt.

4.3. Aggregation Pipeline

After all experiment runs complete, an aggregation script collects results into a single analysis frame. The script iterates over all run directories, loads each `config.yaml` and `metrics.json`, and produces one row per run indexed by model, vignette, slice, and temperature.

Strategy counts are expanded into six separate columns (one per taxonomy category), enabling per-strategy frequency analysis. Alignment scores per strategy are similarly expanded. The resulting DataFrame is saved as a CSV file (`stats/data/all_runs.csv`) that serves as the input for all downstream analysis and visualisation scripts.

This unified frame is a prerequisite for the cross-method analysis in section 5.5. Spearman correlations between Method 1, Method 2, and Method 3 require all three metrics aligned on the same condition index. Without a single frame that maps each condition to its Jaccard, BERTScore, and alignment scores, cross-method comparison would require manual alignment across separate output files.

Downstream scripts in `stats/scripts/` consume this CSV to produce descriptive statistics (`descriptives.py`), statistical tests (`tests.py`), and all five result figures. The separation between aggregation and analysis allows each step to be rerun independently: if a figure script is modified, only visualisation reruns, not the full aggregation.

5. Results

All results are reported at the second rescripting turn (`slice_2`), aggregated across six vignettes. Each data point represents the mean of twenty independent trials. Supplementary figures appear in Appendix C.

5.1. Plan Validity and Strategy Distribution

Before analysing stability, the plan extraction step must be validated. A trial is valid if the model produces a well-formed plan block containing at least one recognised strategy category. Across all 6,000 trials, validity rates exceed 99% for every model. The lowest rates belong to Qwen 3.5 27B (99.3%) and OLMo 3.1 32B (99.7%). All other models achieve 99.8% or above. Validity does not vary meaningfully with temperature. These rates confirm that the fused chain-of-thought prompt reliably elicits structured plan output, and that the subsequent stability metrics are computed over near-complete data (Figure C.1).

Strategy distributions vary across models (Figure 5.1). All ten models draw from the same six-category taxonomy, but the relative frequencies differ. *confrontation* and *self-empowerment* dominate across most models. GPT-5.4 and Sonnet 4.6 show the narrowest distributions, concentrating on two to three categories. Qwen 3.5 27B and Llama 3.3 70B spread selections more evenly across categories. These distributional differences are a prerequisite for interpreting the consistency results that follow. A model that always selects the same two strategies will score high on Jaccard regardless of temperature. A model that draws from a wider repertoire has more room to vary.

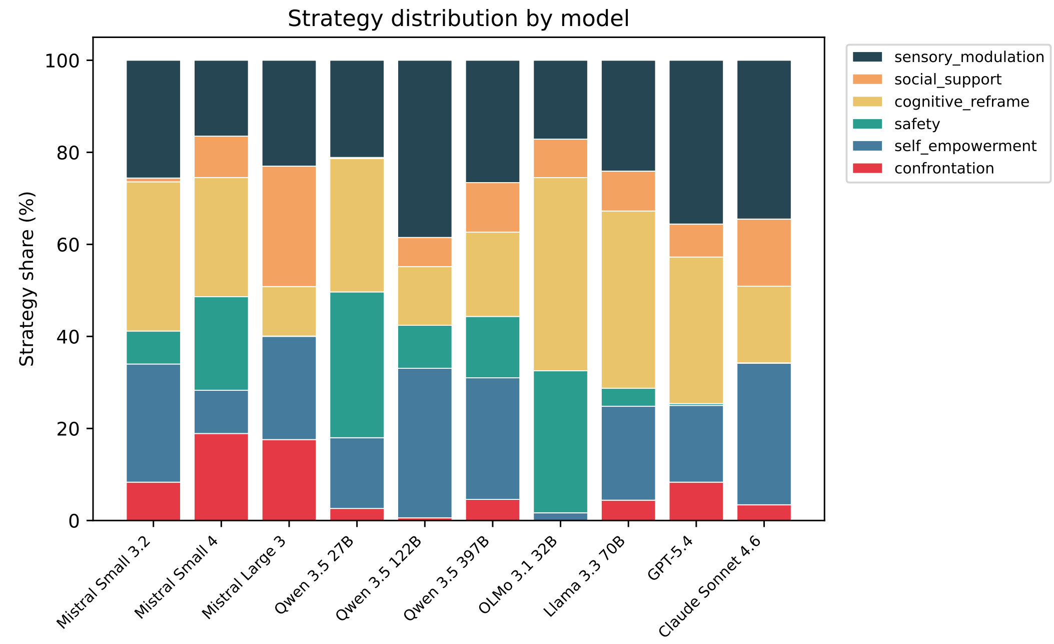

Figure 5.1.: Strategy distribution by model, pooled across temperatures and vignettes. Each bar shows the share of trials selecting each taxonomy category. Models with narrower distributions have less room to vary.

5.2. Plan Consistency (Method 1)

Figure 5.2 presents the headline result: mean Jaccard similarity by model and temperature, grouped by model family.

Modal-set agreement (the proportion of trials selecting the exact same strategy combination) confirms these patterns and is reported in Appendix C.

The two proprietary models, GPT-5.4 and Sonnet 4.6, show the most stable temperature profiles. Their mean Jaccard stays between 0.82 and 0.92 across all five temperature points with no significant trend. Both also show narrower strategy distributions (Figure 5.1), which likely contributes.

Most open-weight models peak at low temperatures. Qwen 3.5 122B and OLMo 3.1 32B reach $J = 1.0$ at $T = 0.0$ but decline at higher settings. The Mistral family behaves differently: all three models peak between $T = 0.075$ and $T = 0.15$ rather than at $T = 0.0$. Mistral Small 3.2 reaches $J = 0.938$ at $T = 0.075$, Mistral Large 3 reaches $J = 0.848$ at $T = 0.15$, and Mistral Small 4 reaches $J = 0.881$ at $T = 0.15$.

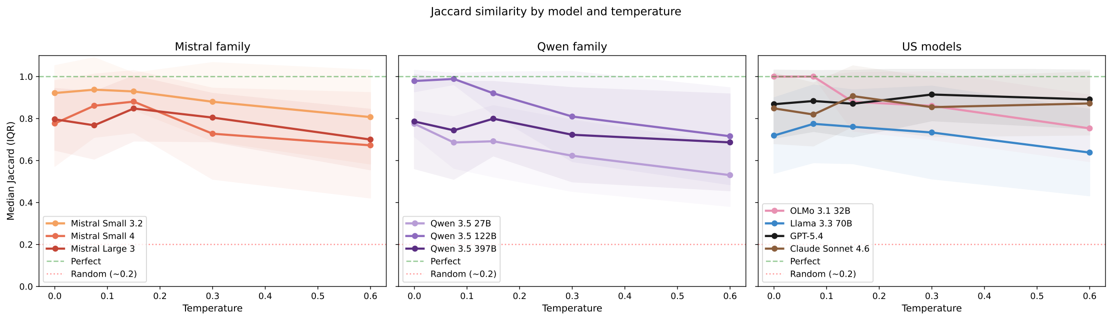

Figure 5.2.: Mean Jaccard similarity by model and temperature, grouped by family. Shaded bands show ±1 SD across vignettes. The dashed line marks the random baseline ($J \approx 0.2$). Higher is more consistent.

This means greedy decoding is not always optimal. Several models score higher at a low non-zero temperature: Mistral Small 4 ($0.777 \rightarrow 0.881$ at $T = 0.15$), Mistral Large 3 ($0.796 \rightarrow 0.848$ at $T = 0.15$), and Llama 3.3 70B ($0.719 \rightarrow 0.775$ at $T = 0.075$). Whether these differences reflect a genuine effect or sample variation ($n = 6$ vignettes per cell) cannot be determined from these results alone.

Temperature sensitivity also varies across models. Per-model Spearman correlations between Jaccard and temperature (Table 5.1) reveal three groups. OLMo 3.1 32B ($\rho = -0.576$, $p = 0.001$), Qwen 3.5 122B ($\rho = -0.560$, $p = 0.001$), and Qwen 3.5 27B ($\rho = -0.455$, $p = 0.012$) show statistically significant degradation. All other models, including the entire Mistral family, show no significant temperature effect on Jaccard. The pooled correlation across all models is $\rho = -0.206$ ($p < 0.001$).

Table 5.1.: Spearman correlation between Jaccard and temperature, per model. Significant correlations ($p < 0.05$) are marked with an asterisk.

Model	ρ	p	Sig.
Mistral Small 3.2	−0.232	0.217	
Mistral Small 4	−0.216	0.251	
Mistral Large 3	−0.081	0.672	
Qwen 3.5 27B	−0.455	0.012	*
Qwen 3.5 122B	−0.560	0.001	*
Qwen 3.5 397B	−0.150	0.427	
OLMo 3.1 32B	−0.576	0.001	*
Llama 3.3 70B	−0.161	0.394	
GPT-5.4	+0.026	0.891	
Claude Sonnet 4.6	+0.028	0.881	
Pooled	−0.206	<0.001	*

5.3. Response Similarity (Method 2)

BERTScore F1 measures semantic similarity between response texts, independent of the plan block. The pooled temperature effect is stronger than for Jaccard ($\rho = -0.467$, $p < 0.001$): rising temperature changes response wording more than it changes strategy selection. However, BERTScore is insensitive to strategy changes: a model that switches from *confrontation* to *cognitive reframe* produces surface text similar enough that BERTScore does not detect the difference. BERTScore therefore serves as a sanity check for response style rather than a measure of decision stability. Full results appear in Appendix C.

5.4. Plan-Response Coherence (Method 3)

Figure 5.3 shows plan-response coherence by temperature and per model. Coherence measures whether the model's response implements the strategies it declared in its plan.

Coherence is temperature-independent. The pooled Spearman correlation between coherence and temperature is $\rho = -0.083$ ($p = 0.154$), not significant. Models do not get worse at implementing their declared strategies as temperature rises. They select *different* strategies more often (lower Jaccard) but implement whichever strategies they select equally well (stable coherence).

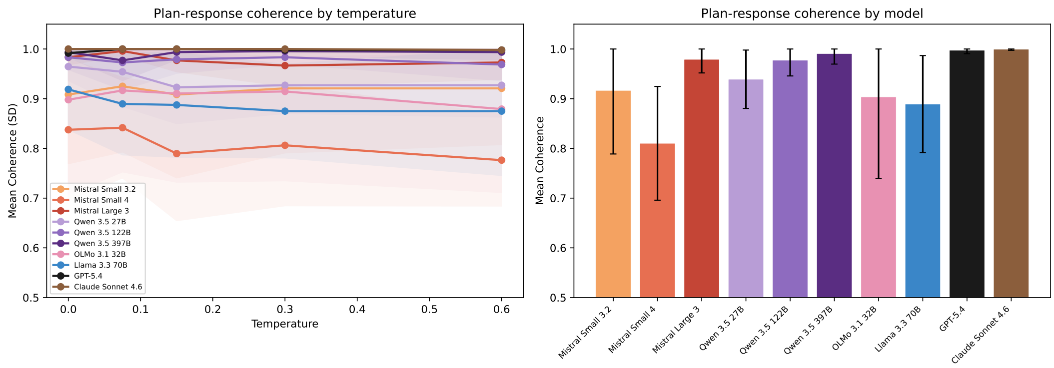

Figure 5.3.: Plan-response coherence by temperature (left) and overall by model (right). Error bars show ±1 SD. Most models implement their declared strategies consistently regardless of temperature.

This separation has a clinical implication: temperature-induced instability occurs at the decision-making level, not at the execution level. A model at high temperature does not produce incoherent responses. It produces coherent responses to a different therapeutic plan.

Kruskal-Wallis tests confirm that models differ significantly on coherence ($p < 0.001$). Mistral Small 4 shows the lowest mean coherence, consistent with its erratic strategy selection at extreme temperatures. The proprietary models and Mistral Small 3.2 cluster near the top.

5.5. Cross-Method Analysis

The three evaluation methods are related but not redundant. Table 5.2 reports Spearman correlations between all method pairs, and Figure 5.4 visualises the relationship between plan consistency and response similarity.

The moderate Jaccard–BERTScore correlation ($\rho = 0.508$) indicates that models with more consistent strategy selection also tend to produce more similar response text, but the relationship is not strong enough for one metric to substitute for the other. The weak Jaccard–Coherence correlation ($\rho = 0.217$) confirms that plan consistency and plan-response coherence measure different properties.

Figure 5.4 shows how the models cluster. GPT-5.4 and Sonnet 4.6 anchor the upper-right corner: high on both dimensions. Qwen 3.5 27B sits in the lower-left: low plan consistency and low response similarity. Mistral Small 4 presents an interesting case: moderate Jaccard but the lowest BERTScore in the set, sug-

Table 5.2.: Spearman correlations between the three evaluation methods. All correlations are significant ($p < 0.001$). The moderate Jaccard–BERTScore correlation and weak Jaccard–Coherence correlation confirm that the methods capture distinct dimensions.

Method pair	ρ	Interpretation
Jaccard vs. BERTScore	0.508	Moderate
Jaccard vs. Coherence	0.217	Weak
BERTScore vs. Coherence	0.393	Weak–moderate

gesting that even when it selects the same strategies, its response wording varies considerably.

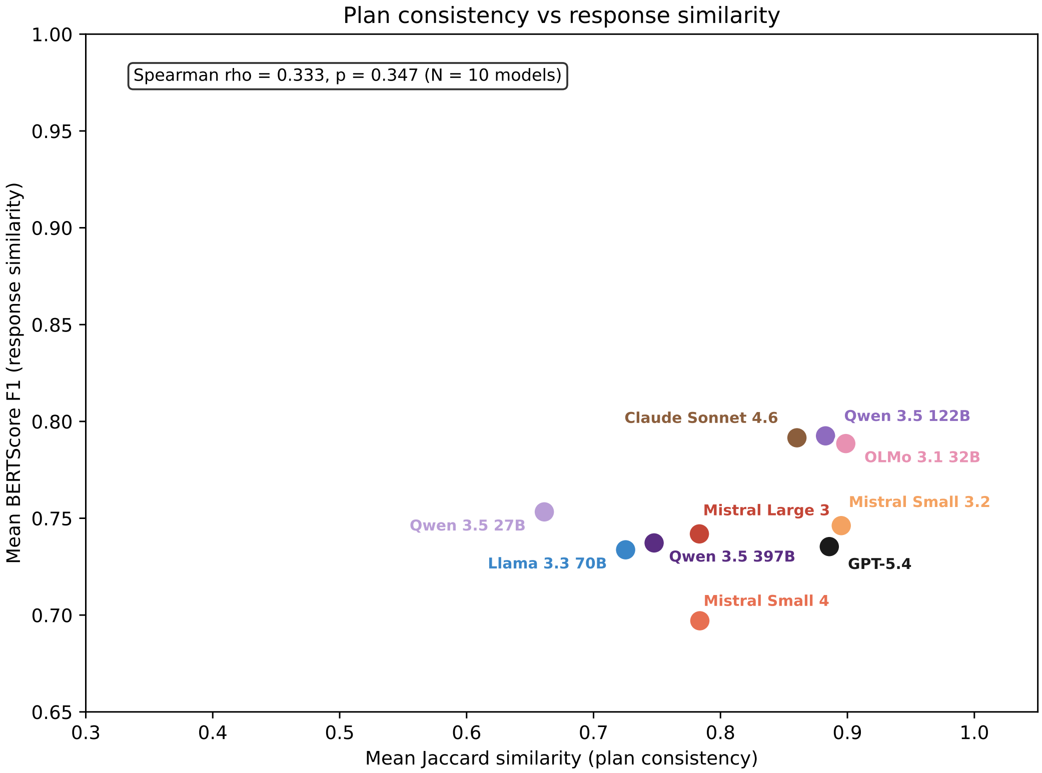

Figure 5.4.: Plan consistency (mean Jaccard) versus response similarity (mean BERTScore F1) per model, pooled across temperatures. The proprietary models cluster in the upper-right corner. Qwen 3.5 27B and Mistral Small 4 occupy opposite ends of the BERTScore axis despite similar Jaccard ranges.

5.6. Secondary Effects

5.6.1. Vignette Effects

Kruskal-Wallis tests show a marginally significant vignette effect on BERTScore ($p = 0.033$) but not on Jaccard ($p = 0.094$). Patient profile affects response wording more than strategy selection. Full per-vignette heatmaps appear in Appendix C.

5.6.2. Conversation Depth

A preliminary analysis tested Mistral Large 3 at five conversation depths (slices 1–5) across all six vignettes at $T = 0.075$ and $T = 0.15$. Figure 5.5 shows the results.

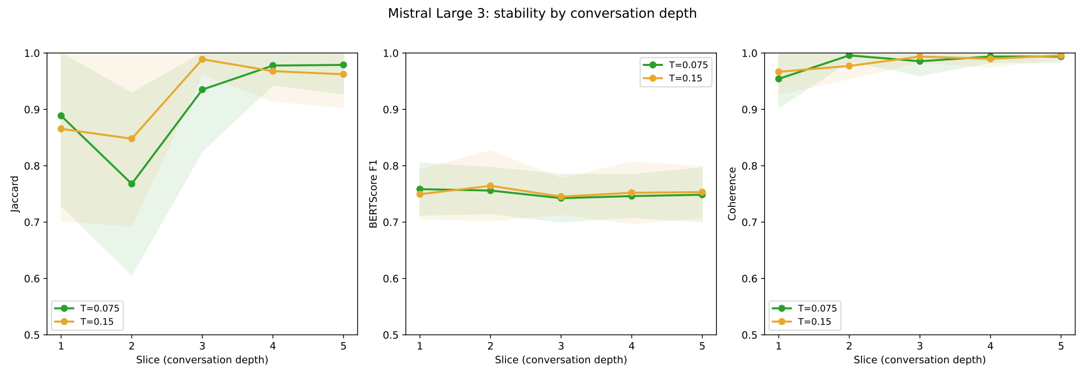

Figure 5.5.: Stability by conversation depth for MISTRAL LARGE 3. Jaccard is lowest at slice 2 and rises to near-ceiling values from slice 3 onward. BERTScore and coherence remain flat across depths.

Jaccard consistency is lowest at the second slice (mean $J = 0.77$ at $T = 0.075$) and rises to near-ceiling values from the third slice onward ($J > 0.93$). BERTScore and coherence remain flat across all depths, confirming that conversation depth affects strategy *selection* but not response *wording* or plan *execution*.

The main experiment evaluates at slice 2, making the reported stability scores a conservative estimate. A model that achieves high consistency at this depth would likely score even higher at later conversation points where prior exchanges provide more anchoring context.

6. Discussion

6.1. Interpreting the Three-Level Framework

The three evaluation methods were designed to capture distinct aspects of therapeutic stability (section 3.7). The results confirm that they do: the cross-method correlations are moderate at best ($\rho = 0.508$ for Jaccard–BERTScore, $\rho = 0.217$ for Jaccard–Coherence). Had any two methods been strongly correlated, one would be redundant. Instead, each method reveals information that the others miss.

The most informative finding is the separation between plan consistency and coherence. Temperature increases reduce Jaccard (models select different strategies) but do not reduce coherence (models implement whichever strategy they select equally well). This pattern localises the source of instability to the *decision-making* layer rather than the *execution* layer. A model at high temperature does not produce incoherent or unfaithful responses. It produces coherent responses to a different therapeutic plan.

This distinction matters in practice. If the model selected consistent strategies but failed to carry them out, the problem would be in the response generation. Instead, the pattern observed here is the opposite: the model carries out its plans faithfully but picks different plans across runs. The instability sits in strategy selection, not in response quality. It also validates Method 1: because plans are faithfully executed, Jaccard over plan sets measures actual decision stability rather than label noise. Temperature tuning is therefore the more promising adjustment, not prompt redesign.

BERTScore occupies a more limited role than initially anticipated. As shown in section 5.3, it cannot distinguish between clinically different plans when the surface text remains similar. This limitation is not a failure of BERTScore itself but a consequence of how therapeutic responses are structured: different rescripting strategies often share the same conversational framing, sentence length, and register. For future work on therapeutic stability, a metric that operates on plan content rather than surface text would be more informative.

6.2. Cross-Model Comparison

Does Mistral Large 3, the EU-sovereign model in this study, achieve therapeutic stability comparable to proprietary alternatives (Figure 5.2)?

At the vendor-recommended temperature ($T = 0.15$), Mistral Large 3 achieves $J = 0.848$, compared to $J = 0.871$ for GPT-5.4 and $J = 0.907$ for Sonnet 4.6. The gap is small. At higher temperatures ($T = 0.6$), Mistral Large 3 drops to $J = 0.700$ while the proprietary models stay above $J = 0.87$. The sovereignty tradeoff is modest at recommended temperatures and grows at higher settings.

Mistral Small 3.2 (24B dense) is the most stable Mistral model ($J = 0.922$–0.938 at low temperatures), though it also degrades at $T = 0.6$ ($J = 0.807$). None of the Mistral models show a statistically significant temperature trend.

Several models score higher at a low non-zero temperature than at $T = 0.0$. Mistral Small 4 rises from $J = 0.777$ at $T = 0.0$ to $J = 0.881$ at $T = 0.15$. Mistral Large 3 follows the same pattern ($0.796 \rightarrow 0.848$). This is consistent with Mistral's optimisation for low non-zero temperatures: the vendor default sits at $T = 0.15$, and the results suggest that stability-sensitive tasks are best served in the range between zero and that default.

Among the open-weight comparators, parameter count does not straightforwardly predict stability. Qwen 3.5 122B ($J = 0.979$ at $T = 0.0$) outperforms Qwen 3.5 397B ($J = 0.786$) at every temperature point. OLMo 3.1 32B achieves $J = 1.0$ at $T = 0.0$ and $T = 0.075$ but degrades significantly at higher temperatures ($\rho = -0.576$, strongest sensitivity in the set).

GPT-5.4 ($J = 0.87$–0.92) and Sonnet 4.6 ($J = 0.82$–0.91) show the most stable temperature profiles, but they are not in a class of their own. Qwen 3.5 122B and Mistral Small 3.2 reach similar ranges. Part of the explanation is distributional: both proprietary models concentrate on a narrow set of strategies (Figure 5.1), leaving less room to vary.

6.3. Temperature and Clinical Deployment

The five-point temperature sweep reveals that the relationship between temperature and stability is not monotonic for all models. As reported in section 5.2, several models achieve higher mean Jaccard at a low non-zero temperature than at $T = 0.0$. These differences are modest, but they are consistent with Mistral's vendor default of $T = 0.15$: stability-sensitive tasks may be best served in the range between zero and the default.

The pooled Spearman correlation ($\rho = -0.206$, $p < 0.001$, Table 5.1) confirms that higher temperature generally reduces plan consistency, but the effect is weak. Per-model analysis reveals that only three models show statistically significant degradation: OLMo 3.1 32B, Qwen 3.5 27B, and Qwen 3.5 122B. The remaining seven models, including the entire Mistral family, show no significant temperature effect on Jaccard.

For clinical deployment, these results suggest a practical recommendation: operate at or near the vendor-recommended temperature. Models designed for a specific temperature range tend to be most stable within it. Pushing a model to $T = 0.0$ in the hope of maximising determinism can be counterproductive.

Coherence is temperature-independent across all models ($\rho = -0.083$, $p = 0.154$). This means that the clinical cost of higher temperature is limited to strategy *selection* variability. The model does not become worse at *executing* its chosen strategy. A patient receiving care from a high-temperature model would encounter different therapeutic approaches across sessions, but each individual session would be internally coherent.

6.4. Vignette-Dependent Behaviour

The vignette effect on strategy consistency is not statistically significant ($p = 0.094$), though BERTScore shows a marginally significant effect ($p = 0.033$): some patient profiles produce more variable response wording than others. The per-vignette heatmaps in Appendix C show that individual model-vignette combinations can differ noticeably from the aggregate, even if the overall effect is weak. Per-vignette reporting remains useful for identifying conditions under which a specific model's consistency breaks down.

6.5. Evaluation Framework Reflections

Measuring at three layers revealed a pattern no single metric could show: strategy selection varies while execution stays faithful. BERTScore alone would have missed this, since it captures therapeutic tone but not the underlying strategy choices (section 5.3). Frozen histories and 20-trial repetition (Blair-Stanek & Van Durme, 2025) ensure that observed variance comes from the model, not from differences in input or sample size.

That said, the consistency scores depend on how strategies are categorised. Broader categories inflate Jaccard (more selections land in the same bin), finer categories deflate it. The six-category taxonomy (Table 3.1) balances clinical detail with the ability to tell strategies apart, but different category boundaries would shift the absolute numbers. The relative ordering of models is more stable, since all are evaluated against the same set.

The plan block also has limits. It captures what the model declares, not what it internally computes (section 3.5). It cannot tell apart a model that picks a strategy and then writes the response from one that writes a response and then picks a matching label. The high coherence scores fit either reading. What matters is that declarations vary across runs, and that variation predicts variable therapeutic behaviour regardless of the underlying process.

Beyond IRT, other structured therapy protocols where the therapist chooses from a defined set of techniques could be evaluated the same way. CBT and motivational interviewing are likely candidates. Each would need its own strategy taxonomy, but the evaluation setup and metrics carry over.

Two important caveats apply. First, this framework measures whether a model produces *consistent* responses, not whether those responses are *clinically appropriate*. A model that consistently selects the wrong strategy scores high on plan consistency. Stability is necessary for clinical deployment but not sufficient. All evaluations use synthetic patients and no human participants are involved.

Second, no reproducibility seed was set, as not all vendors support one. Even where seeds are available, they provide best-effort determinism at most (Yuan et al., 2025). A supplementary seed experiment on Mistral Large 3 confirms this: setting a fixed seed (`seed=42`) does not improve consistency compared to unseeded runs (Figure C.6). The trial repetition design captures stochastic variance directly.

The experiment evaluates at a single conversation depth (`slice_2`), chosen as a conservative estimate based on the preliminary depth analysis (section 3.3). Results at other depths may differ. All models were accessed through OpenRouter with fixed inference providers, but provider-side batching and load balancing may still contribute to the non-determinism observed at $T = 0.0$.

Further limitations regarding scope, vignette coverage, judge validation, and taxonomy constraints are discussed in section 7.2.

7. Conclusion

7.1. Summary of Findings

This thesis asked how consistent LLM therapeutic responses are across independent runs within a structured IRT protocol. Three evaluation methods were applied to ten models across 6,000 trials spanning five temperature conditions.

The main findings are:

1. **Consistency varies widely across models and temperatures.** Mean Jaccard ranges from $J = 0.56$ (Qwen 3.5 27B at $T = 0.6$) to $J = 1.0$ (OLMo 3.1 32B at $T = 0.0$). Temperature is the primary driver of variation for open-weight models.

2. **The EU-sovereign model is close to proprietary ones models at recommended temperatures.** Mistral Large 3 achieves $J = 0.848$ at $T = 0.15$, compared to $J = 0.871$ for GPT-5.4 and $J = 0.907$ for Sonnet 4.6. The gap widens at higher temperatures.

3. **Instability sits in strategy selection, not execution.** Temperature reduces plan consistency but not plan-response coherence. Models pick different strategies at higher temperatures but carry out whichever strategy they pick equally well.

4. **Greedy decoding is not always optimal.** Three models achieve higher consistency at $T = 0.075$ than at $T = 0.0$. Whether this reflects a genuine benefit of low sampling variance or an artefact of the 20-trial sample size cannot be determined from these results alone.

5. **No single metric captures the full picture.** The three evaluation levels are only moderately correlated ($\rho = 0.217$ to 0.508). BERTScore misses strategy changes.

6. **Patient profile may matter.** The vignette effect on Jaccard is not significant ($p = 0.094$), but BERTScore shows a marginal effect ($p = 0.033$). Per-vignette reporting can reveal weaknesses that aggregate scores mask.

Regarding the two contributions stated in section 1.4: the three-level framework (Contribution 1) fulfilled its design goal by identifying that instability is concentrated at the strategy selection layer. Neither BERTScore nor coherence alone would have been enough to revealed that. The cross-model comparison (Contribution 2) shows that Mistral Large 3 approaches proprietary-level stability at its recommended operating temperature, though a small gap remains.

7.2. Limitations

The study evaluates a single phase (rescripting) of a single protocol (IRT). The rescripting phase was selected because it requires active strategy decisions, but stability in other stages or other therapeutic protocols may differ.

All patient interactions are simulated. Synthetic patients follow scripted behavioural profiles and do not exhibit the unpredictable responses of real patients. Six profiles sample the difficulty range (section 3.2) but cannot claim exhaustive coverage. No human participants were involved in this study, and the results measure computational consistency rather than clinical safety or efficacy. IRT targets nightmares rather than psychiatric diagnoses, which limits clinical risk, but any deployment to real patients would require validation beyond what this framework provides.

The LLM judge (Gemini 3.1 Pro) was validated on 60 stratified trials with substantial human agreement ($\kappa = 0.777$), but this covers only 1% of the 6,000 total trials. A larger annotation study would strengthen confidence in the coherence scores.

The six-category strategy taxonomy constrains what can be measured. Strategies that fall outside these categories are invisible to Method 1, and different category boundaries would produce different absolute consistency scores.

Finally, consistency is not the same as quality. A model that consistently selects an inappropriate strategy scores high on plan consistency. Stability is necessary for clinical deployment but not sufficient on its own.

7.3. Future Work

Extending the evaluation beyond the rescripting phase to all IRT stages would test whether the observed patterns hold when the model faces different types of clinical decisions. More broadly, the three-level framework applies to other therapy protocols where the therapist chooses from a defined set of techniques, such as CBT or motivational interviewing. Each would need its own strategy taxonomy, but the evaluation setup carries over.

Since model vendors update weights and infrastructure without notice, repeating the evaluation across model versions would show whether stability improves or fluctuates over time. reDreamAI targets German-speaking users, so evaluating stability in German would test whether the English-language patterns transfer.

Finally, newer model architectures may affect output consistency independently of temperature. Whether architectural choices such as hybrid attention mechanisms can improve therapeutic consistency beyond what temperature tuning achieves remains an open question.

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Overview of Tools Used

In accordance with the AI Tools Guidelines of the University of Osnabrück, the following describes the supporting role of LLM-based tools in the research and writing of this thesis.

For intellectual engagement and brainstorming, I used Claude 4.6 (Anthropic)[1] and Gemini as discussion partners. These tools were used to challenge my reasoning and suggest alternative framings for developed arguments. As English is not my first language, I occasionally used these tools to refine phrasing where my original formulations lacked fluency or precision.

For software development and data processing, I briefly used Cursor to develop the evaluation pipeline. My primary tool was Claude Code for data visualisation, debugging and LaTeX formatting. I challenged and verified all AI-generated output before inclusion.

All arguments, interpretations, and conclusions are my own.

[1] https://claude.ai

Glossary

API Application Programming Interface

CBT Cognitive Behavioral Therapy

CoT Chain-of-Thought

EU European Union

IRT Imagery Rehearsal Therapy

LLM Large Language Model

MoE Mixture of Experts

NLI Natural Language Inference

NLP Natural Language Processing

PTSD Post-Traumatic Stress Disorder

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A. Strategy Taxonomy

The final six-category taxonomy (v4) constrains the `<plan>` block to the following therapeutic rescripting mechanisms. Each category describes *how* the nightmare changes, not what outcome it achieves. The taxonomy evolved through four iterations to resolve distribution skew (section 3.6).

Table A.1.: Strategy taxonomy v4: full definitions as provided to the therapist model. NOT-examples reduce category overlap.

Category	Definition (as given to the model)
confrontation	The dreamer directly faces, fights, or stands up to the threat itself (NOT avoiding, escaping, or gaining general confidence).
self-empowerment	The dreamer gains a new ability, skill, or inner strength they did not have before (e. g. discovers they can fly, becomes stronger, unlocks a hidden power). The DREAMER changes, not the threat. (NOT reinterpreting existing elements, which is *cognitive reframe*.)
safety	A physical barrier, shelter, or protective figure is added to shield the dreamer (e. g. walls, locked doors, a guardian). NOT just feeling calm or changing the scene.
cognitive reframe	The meaning or identity of existing dream elements changes (e. g. a monster becomes harmless, an enemy becomes a friend). The THREAT changes, not the dreamer. (NOT the dreamer gaining new powers, which is *self-empowerment*.)
social support	People in the dream become supportive, or a new ally is introduced. Existing characters can shift from threatening to encouraging. Any form of interpersonal support counts.
sensory modulation	The dream's sensory qualities change: what is seen, heard, or felt shifts (e. g. darkness becomes light, harsh sounds become soft music, the dreamer feels calm through breathing or self-talk). Covers both external environment changes and internal somatic calming.

Revision history:

v1 (8 categories): Included *empowerment* and *mastery* as separate categories. These two consumed 87.7% of all strategy picks across 1,272 trials, while three categories were never selected. Root cause: both describe a therapeutic *goal* (the patient gains control) rather than a *mechanism*.

v2 (7 categories): Merged *empowerment* and *mastery* into *agency*. The merged category achieved 100% dominance.

v3 (7 categories): Split *agency* along the mechanism dimension into *confrontation* (external action) and *self-empowerment* (internal transformation). Sharpened all descriptions to focus on mechanism, not outcome.

v4 (6 categories): Merged *emotional regulation* into *sensory modulation*. In smoke testing (120 trials), *emotional regulation* appeared once (0.4%). In single-turn rescripting, internal calming manifests as sensory change.

B. Prompts

This appendix reproduces the key prompts used in the evaluation pipeline. All prompts are stored as YAML files in `data/prompts/`.

B.1. Fused Plan and Response Prompt

The following system prompt is injected into the therapist model during evaluation. The `{categories_block}` placeholder is replaced with the full taxonomy definitions from Table A.1.

```
You are an Imagery Rehearsal Therapy (IRT) assistant
helping a patient rescript their nightmare.

First, select 1-2 therapeutic strategies from this taxonomy:
{categories_block}

Choose the strategies most specifically suited to THIS
patient's nightmare content. Avoid defaulting to general
approaches when a more specific mechanism fits.

Then respond to the patient using those strategies.

STRICT output format:
Line 1: <plan>category1 / category2</plan>
Line 2: (empty)
Line 3+: Your therapeutic response (conversational,
         supportive, max 3 sentences)
```

B.2. Coherence Judge Prompt

The following prompt is sent to Gemini 3.1 Pro ($T = 1.0$, thinking enabled). The `{taxonomy_block}` is replaced with the full strategy definitions. The `{strategies_block}` and `{response}` are filled per trial.

```
System:
You are evaluating whether a therapist’s response implements
specific therapeutic strategies from IRT.

## Strategy Definitions
{taxonomy_block}

## Scoring Rubric
For each declared strategy, assign ONE score:
- 2 (implemented): The response clearly and specifically
  implements this strategy.
- 1 (partial): The response touches on this strategy but
  does not fully develop it.
- 0 (absent): The response does not implement this strategy.

## Instructions
1. Read the therapist response carefully.
2. For each declared strategy, provide brief reasoning
   (1-2 sentences), then your score.
3. Use EXACTLY this output format for each strategy:

strategy_id: <reasoning> | score: <0|1|2>

User:
## Declared Strategies
{strategies_block}

## Therapist Response
{response}
```

B.3. Example Patient Vignette

Six vignettes define patient profiles for dialogue generation. The following example shows the *anxious* profile. All vignettes follow the same schema.

```
{
  "type": "anxious",
  "name": "Maya",
  "age": 24,
  "gender": "female",
  "background": "A graduate student dealing with nightmares
```

```
    that started during a particularly stressful thesis
    period. Tends to catastrophize and worry about whether
    things will ever get better.",
  "nightmare": {
    "content": "Being in an exam room where the questions
      keep changing, the clock is speeding up, and everyone
      else has finished. Sometimes the room starts flooding
      or the walls close in.",
    "frequency": "5-6 times per week",
    "duration": "8 months"
  },
  "personality_traits": [
    "worried", "seeking reassurance",
    "catastrophizing", "hyperaware of sensations",
    "needs frequent validation"
  ],
  "resistance_level": "moderate",
  "sample_responses": [
    "What if I try to change the dream and it makes it
     worse? Has that ever happened?",
    "I'm already anxious just thinking about the nightmare.
     I don't know if I can do this."
  ]
}
```

C. Supplementary Results

This appendix contains figures that support the main text but were deferred to keep the results chapter focused on the primary findings.

C.1. Plan Validity Rate

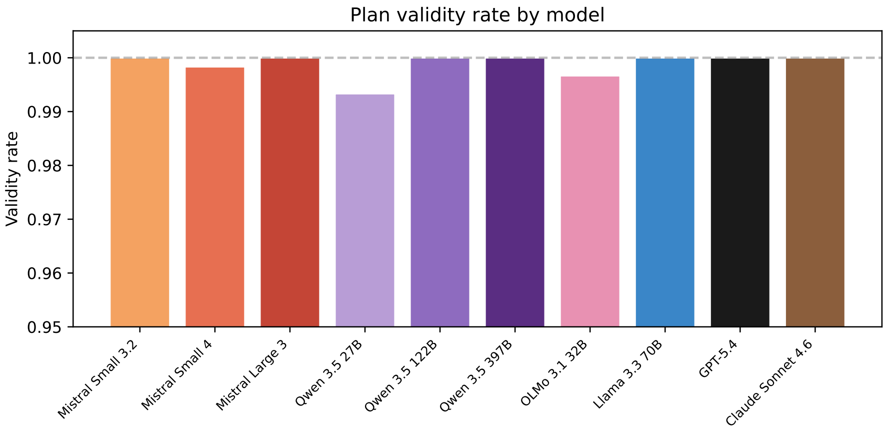

Figure C.1.: Plan validity rate by model, pooled across temperatures and vignettes. All models exceed 99%. The y-axis begins at 0.95 to show the small differences between models.

C.2. Modal-Set Agreement

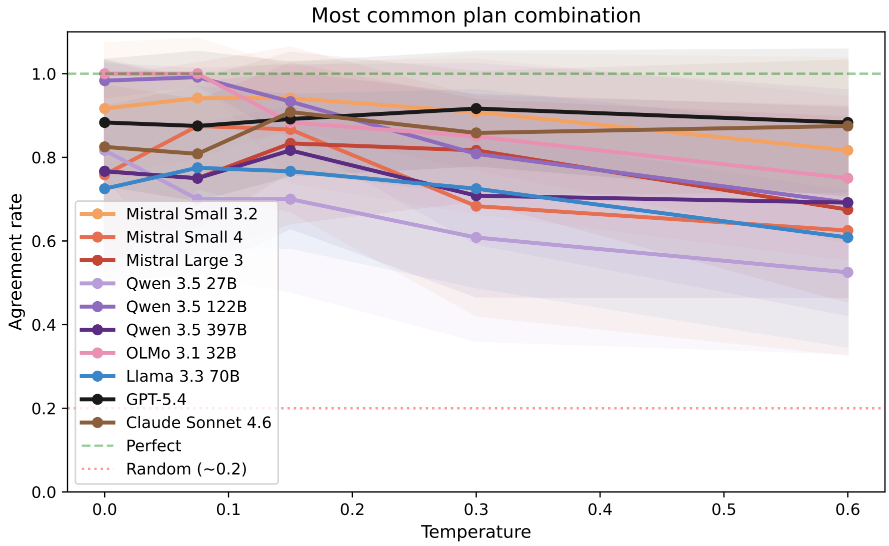

Figure C.2.: Modal-set agreement by model and temperature. The agreement rate is the proportion of trials selecting the most common strategy combination per condition.

C.3. BERTScore by Model and Temperature

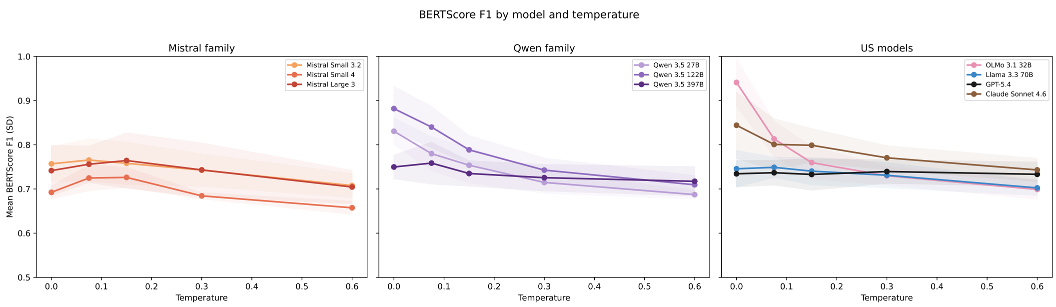

Figure C.3.: Mean BERTScore F1 by model and temperature, grouped by family. The pooled temperature effect is stronger than for Jaccard ($\rho = -0.467$), but BERTScore is insensitive to strategy differences (section 5.3).

C.4. Vignette Heatmaps

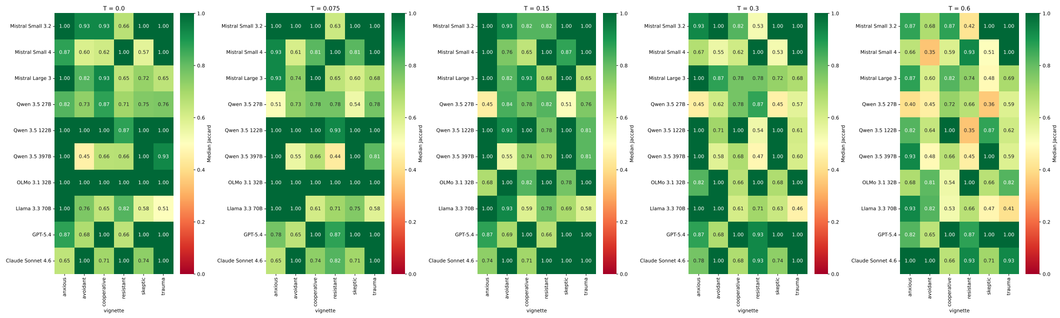

Figure C.4.: Jaccard by model and vignette, one panel per temperature. Some vignettes show lower consistency for individual models, though the overall vignette effect on Jaccard is not statistically significant ($p = 0.094$).

C.5. Slice Depth Heatmap

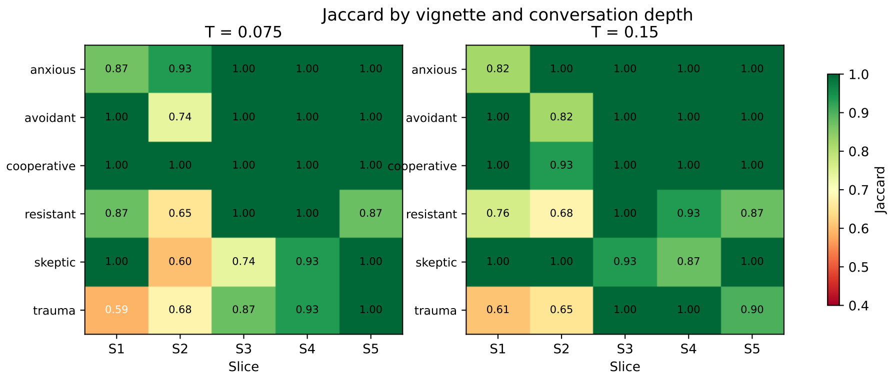

Figure C.5.: Jaccard by vignette and conversation depth for Mistral Large 3, one panel per temperature. Slice 2 shows the lowest consistency across most vignettes, supporting its selection as a conservative evaluation point.

C.6. Seed Comparison

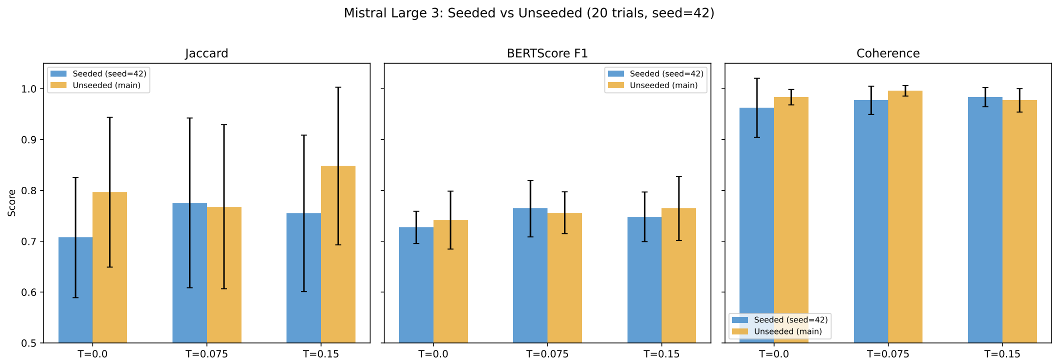

Figure C.6.: Seed comparison for Mistral Large 3. Setting a fixed seed (`seed=42`) does not improve Jaccard, BERTScore, or coherence compared to unseeded runs. Error bars show ± 1 SD across vignettes. Differences are within noise.